

RAG Validation Test Suite

Test Category 1: Cross-Domain Isolation

- **Objective:** Verify that the vector database handles mismatched multi-file contexts correctly and that the LLM does not hallucinate false business relationships.
- **Test Scenario:** Pass two completely independent domains (e.g., `employee_records.sql` and `product_catalog.csv`).
- **Expected Result:** The system should explicitly state that there is no semantic overlap or valid key constraint joining the datasets.
- **Actual Result (Observed):** System correctly returned a negative integration response. **(Pass)**

Pipeline Architecture

Production configuration state used for runtime context injections.

LLM Engine:

gpt-4o

Vector Store Tiering:

Hybrid Retrieval Vector Engine
Active (Ensemble: 60%
Corporate KB / 40%
Operational Schemas)

Ingest Data Entities

akdemocsvfile.csv
1,200

postgrepractice.sql
1,200

GenAI Agent for Data Pipeline Assistance

Automated structural schema extraction and rule-compliance evaluation

guide me through these uploaded files. The data in akdemofile.csv is related to different products; and data in postgrepractice.sql is about employees. Tell if these both can be integrated in any way for my e-commerce business use-case

Based on the uploaded content files and data:

Analysis of Provided Data:

akdemocsvfile.csv

This dataset appears to contain transaction/product-related data for an e-commerce business.
Columns represent details such as:
Transaction ID (536378 , 536387 , etc.)
Product Code (22386 , 23929 , etc.)
Product Description (30MB0 BAG PSN POLKADOT , etc.)
Quantity Sold (18 , 8 , etc.)
Transaction Date-Time (2025-12-01 08:34 , etc.)
Price Per Unit (e.g., 1.95 , 2.18 , etc.)
Customer ID (14680 , 13047 , etc.)
Customer Location (e.g., United Kingdom , France)
This is clearly linked to sales/transactions in an e-commerce use case, containing information about the products, sales quantities, customers, and locations.

postgrepractice.sql

Tables:
Table emp : Contains employee data and has the following columns:
emp_id , first_name , last_name , gender , birthdate , email , and salary .
Purpose: Employee management for the organization hiring these entities.
Table accounts : Contains account details, with:

Table accounts : Contains account details, with:
id , name , and balance .
Purpose: Likely a financial system for users/customers or internal accounts.

Integration Analysis:

The two datasets serve different purposes and entities, and there's no direct logical link between them at the current context:

akdemocsvfile.csv is focused on external-facing (customers and product transactions) data related to an e-commerce platform.
postgrepractice.sql (Tables emp and accounts) contains internal employee and possibly financial accounts data, which is not directly tied to the customers or products in akdemocsvfile.csv .
The only potential overlap could be if your employees (in the emp table) also acted as sales agents or managers managing relationships with customers (linked to customer_id in akdemocsvfile.csv). However, such a link is not present or identifiable in the current context data.

Recommendation:

If the goal is to build an integrated pipeline for an e-commerce business:

Use the akdemocsvfile.csv file to create tables specifically for transactions, customers, and products:

ecommerce_transactions (transaction_id,product_id,quantity,transaction_date,unit_price,customer_id,country)
products ([product_id,product_name,product_price])
customers ([customer_id,country,etc])

Use the postgrepractice.sql -> emp table for HR/employee management purposes.
If there is a solid business justification (e.g., assigning each transaction to sales representatives from the emp table), you'll need to introduce a foreign key in the e-commerce sales tables to associate transactions/customers with employees.

Conclusion:

The two datasets cannot be directly integrated given the current format of the data. They serve distinct use cases:

akdemocsvfile.csv is for e-commerce transactions/products/customers.
postgrepractice.sql is for employee and account management. The integration would require a conceptual connection in your business model, such as employees acting as sales representatives or account managers for customers. In this scenario, additional columns like sales_rep_id or manager_id (foreign keys referencing the emp_id column in emp) could be included in the e-commerce tables.

Explored Context Sources

Explored Context Sources

Corporate Policy Document:

2. Table and Column Naming Schema ***Snake Case Only***: All object names must be strictly lowercase using underscores (e.g., "customer_orders", not "CustomerOrders" or "customerOrders"). **Timestamp Suffixes**: Every pipeline must append standard metadata fields to target tables: " ...

File: file:postgrepractise.sql

File: postgrepractise.sql | Table emp [[{"cid": 0, "name": "emp_HF", "type": "int"}, {"cid": 1, "name": "first_name", "type": "varchar(50)"}, {"cid": 2, "name": "last_name", "type": "varchar(50)"}, {"cid": 3, "name": "gender", "type": "char(2)"}, {"cid": 4, "name": "birthdate", "type": "date"}, {"cid": ...

File: file:akdemocsvfile.csv

File: akdemocsvfile.csv | 536378 | JUMBO BAG PINK POLKADOT | 10 | 2025-12-01 09:37 | 1.95 | 14688 | United Kingdom | File: akdemocsvfile.csv | 536378 | 21929 | JUMBO BAG PINK VINTAGE PANSLEY | 10 | 2025-12-01 09:37 | 1.95 | 14688 | United Kingdom...

File: file:postgrepractise.sql

File: postgrepractise.sql | Table accounts [{"cid": 0, "name": "id", "type": "int"}, {"cid": 1, "name": "name", "type": "varchar(100)"}, {"cid": 2, "name": "balance", "type": "dec(15,2)"}]...

File: file:akdemocsvfile.csv

File: akdemocsvfile.csv | 536367 | 22745 | POPPY'S PLAYHOUSE BEDROOM | 6 | 2025-12-01 08:34 | 2.1 | 13047 | United Kingdom | File: akdemocsvfile.csv | 536367 | 22748 | POPPY'S PLAYHOUSE KITCHEN | 6 | 2025-12-01 08:34 | 2.1 | 13047 | United Kingdom | File: akdemocsvfile.csv | 536368 | 22960 | JAM MAKI...

File: file:akdemocsvfile.csv

File: akdemocsvfile.csv | 536369 | 21756 | BATH BUILDING BLOCK WORD | 3 | 2025-12-01 08:35 | 5.95 | 13947 | United Kingdom | File: akdemocsvfile.csv | 536370 | 22728 | ALARM CLOCK BAKELIKE PINK | 24 | 2025-12-01 08:45 | 3.75 | 12583 | France | File: akdemocsvfile.csv | 536370 | 22727 | ALARM CLOCK BA...

Test Category 2: Ask an out-of-domain general knowledge question

Expected Result: The model must output the exact fallback string: "I cannot find the answer to this question in the uploaded files or knowledge base."

who is the prime minister of India?

I cannot find the answer to this question in the uploaded files or knowledge base.

Explored Context Sources

Corporate Policy Document:
Corporate Data Pipeline Standards & Naming Conventions This document outlines the mandatory architectural standards, naming schemas, and design constraints for all ingestion pipelines across enterprise data platforms...

File: aksdemocsvfile.csv
File: aksdemocsvfile.csv | 536366 | 22633 | HAND WARMER UNION JACK | 6 | 2025-12-01 08:28 | 1.85 | 17850 | United Kingdom File: aksdemocsvfile.csv | 536366 | 22632 | HAND WARMER RED POLKA DOT | 6 | 2025-12-01 08:28 | 1.85 | 17850 | United Kingdom File: aksdemocsvfile.csv | 536367 | 84879 | ASSORTED ...

File: aksdemocsvfile.csv
File: aksdemocsvfile.csv | 536375 | 851234 | WHITE HANGING HEART T-LIGHT HOLDER | 6 | 2025-12-01 09:32 | 2.55 | 17850 | United Kingdom File: aksdemocsvfile.csv | 536376 | 22114 | HOT WATER BOTTLE TEA AND SYRUP | 48 | 2025-12-01 09:32 | 3.45 | 15291 | United Kingdom File: aksdemocsvfile.csv | 536377 ...

File: postgrepractice.sql
File: postgrepractice.sql | Table emp: [{"cid": 0, "name": "emp_id", "type": "int"}, {"cid": 1, "name": "first_name", "type": "varchar(50)"}, {"cid": 2, "name": "last_name", "type": "varchar(50)"}, {"cid": 3, "name": "gender", "type": "char(2)"}, {"cid": 4, "name": "birthdate", "type": "date"}, {"cid": 5, "name": "email", "type": "varchar(100)"}, {"cid": 6, "name": "salary", "type": "decimal(18,4)"}, {"cid": 7, "name": "inserted_at_utc", "type": "timestamp"}, {"cid": 8, "name": "updated_at_utc", "type": "timestamp"}]

File: aksdemocsvfile.csv
File: aksdemocsvfile.csv | 536370 | 21724 | PANDA AND BUNNY STICKER SHEET | 12 | 2025-12-01 08:45 | 0.85 | 12583 | France File: aksdemocsvfile.csv | 536371 | 22086 | PAPER CHAIN KIT 50'S CHRISTMAS | 80 | 2025-12-01 09:00 | 2.55 | 13748 | United Kingdom File: aksdemocsvfile.csv | 536372 | 22632 | HAN...

File: aksdemocsvfile.csv
File: aksdemocsvfile.csv | 536373 | 851234 | WHITE HANGING HEART T-LIGHT HOLDER | 6 | 2025-12-01 09:02 | 2.55 | 17850 | United Kingdom File: aksdemocsvfile.csv | 536373 | 71053 | WHITE METAL LANTERN | 6 | 2025-12-01 09:02 | 3.39 | 17850 | United Kingdom File: aksdemocsvfile.csv | 536374 | 21258 | Vt...

Test Category 3: Upload two files and queries from one of the file.

Expected Result: The model must consider only that file input.

Pipeline Architecture

Production configuration state used for runtime context injections.

LLM Engine:
gpt-4o

Vector Store Tiering:
Hybrid Retrieval Vector Engine
Active (Ensemble: 60%
Corporate KB / 40%
Operational Schemas)

Ingest Data Entities
aksdemocsvfile.csv 2.2KB
postgrepractice.sql 4.2KB

what are the primary and foreign keys in the uploaded sql file and provide the recommended data model for this

Analysis of Primary and Foreign Keys:

Based on the provided SQL file schema details:

emp Table

Primary Key: The emp_id column (int)
Suggested change: Convert emp_id to bigint or uuid (preferred) since the pipeline standard specifies using UUID or auto-incrementing bigint for primary keys.
Foreign Keys: No foreign keys are defined in this table.

accounts Table

Primary Key: The id column (int)
Suggested change: Convert id to bigint or uuid (preferred) for consistency with pipeline standards.
Foreign Keys: No foreign keys are defined in this table.

Recommended Data Model:

Based on the schemas provided, below is the suggested data model, normalized and adhering to the stated pipeline design standards:

Table: emp

Column Name	Data Type	Constraint	Description
emp_id	UUID	Primary Key	Unique identifier for an employee.
first_name	VARCHAR(50)	NOT NULL	Employee's first name.
last_name	VARCHAR(50)	NOT NULL	Employee's last name.
gender	CHAR(2)	CHECK (gender = 'M' OR gender = 'F')	Employee's gender.

Based on the schemas provided, below is the suggested data model, normalized and adhering to the stated pipeline design standards:

Table: emp

Column Name	Data Type	Constraint	Description
emp_id	UUID	Primary Key	Unique identifier for an employee.
first_name	VARCHAR(50)	NOT NULL	Employee's first name.
last_name	VARCHAR(50)	NOT NULL	Employee's last name.
gender	CHAR(2)	CHECK (gender = 'M' OR gender = 'F')	Employee's gender.
birthdate	DATE	NOT NULL	Employee's date of birth.
email	VARCHAR(100)	UNIQUE	Employee's email address.
salary	DECIMAL(18,4)		Employee's salary.
inserted_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record insert timestamp.
updated_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last record update timestamp.

Table: accounts

Column Name	Data Type	Constraint	Description
id	UUID	Primary Key	Unique identifier for an account.
name	VARCHAR(100)	NOT NULL	Name of the account.
balance	DECIMAL(18,4)	DEFAULT 0.00	Account balance.
inserted_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record insert timestamp.
updated_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last record update timestamp.

Suggested Relationships Between Tables:

Pipeline Architecture

Production configuration state used for runtime context injections.

LLM Engine:

gpt-4o

Vector Store Tiering:

Hybrid Retrieval Vector Engine

Active (Ensemble: 60% Corporate KB / 40% Operational Schemas)

Ingest Data Entities

aksdemocsvfile.csv

1.2KB

postgrespractice.sql

4.2KB

+

Suggested Relationships Between Tables:

Although your uploaded data does not explicitly define relationships, here are some recommended foreign key relationships in a real-world scenario:

Link employees to accounts:
Add a `created_by_emp_id` column (foreign key) to the `accounts` table.
This links the `emp_id` from the `emp` table to the `accounts` table, representing which employee created the account.

Updated `accounts` Table:

Column Name	Data Type	Constraint	Description
id	UUID	Primary Key	Unique identifier for an account.
name	VARCHAR(100)	NOT NULL	Name of the account.
balance	DECIMAL(18,4)	DEFAULT 0.00	Account balance.
created_by_emp_id	UUID	Foreign Key -> emp.emp_id	Employee who created the account.
inserted_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record insert timestamp.
updated_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last record update timestamp.

Summary of Recommendations:

Standardize primary keys to `UUID` or `BIGINT` as per pipeline standards.
Use `DECIMAL(18,4)` for monetary amounts (`salary` and `balance`) to handle financial calculations accurately.
Add the required metadata columns: `inserted_at_utc` and `updated_at_utc` to all target tables.
Clearly define relationships using foreign keys, e.g., linking `emp` and `accounts` .
This normalized data model improves data integrity, adheres to the given pipeline standards, and prepares the tables for scalable and reliable data pipelines.

Test category 4: Ask different questions. Simulating as different users and session-based answers.

Expected Result: Give answers to what is specifically uploaded by user.

Data Pipeline Architect

localhost:8501

Deploy

Pipeline Architecture

Production configuration state used for runtime context injections.

LLM Engine:

gpt-4o

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Hybrid Retrieval Vector Engine

Active (Ensemble: 60% Corporate KB / 40% Operational Schemas)

Ingest Data Entities

aksdemocsvfile.csv

1.2KB

postgrespractice.sql

4.2KB

+

id	UUID	Primary Key	Identifier for an account.
name	VARCHAR(100)	NOT NULL	Name of the account.
balance	DECIMAL(18,4)	DEFAULT 0.00	Account balance.
created_by_emp_id	UUID	Foreign Key -> emp.emp_id	Employee who created the account.
inserted_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record insert timestamp.
updated_at_utc	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last record update timestamp.

Summary of Recommendations:

Standardize primary keys to `UUID` or `BIGINT` as per pipeline standards.
Use `DECIMAL(18,4)` for monetary amounts (`salary` and `balance`) to handle financial calculations accurately.
Add the required metadata columns: `inserted_at_utc` and `updated_at_utc` to all target tables.
Clearly define relationships using foreign keys, e.g., linking `emp` and `accounts` .
This normalized data model improves data integrity, adheres to the given pipeline standards, and prepares the tables for scalable and reliable data pipelines.

Explored Context Sources

Ask a design question about your data schemas...

Data Pipeline Architect

localhost:8501

Deploy

Pipeline Architecture

Production configuration state used for runtime context injections.

LLM Engine:

gpt-4o

Vector Store Tiering:

Hybrid Retrieval Vector Engine

Active (Ensemble: 60% Corporate KB / 40% Operational Schemas)

Ingest Data Entities

aksdemocsvfile.csv

1.2KB

+

GenAI Agent for Data Pipeline Assistance

Automated structural schema extraction and rule-compliance evaluation

what are all the datatypes that are to be considered?

To identify all the necessary data types for the provided dataset, we first analyze each column of the structured records in the file `aksdemocsvfile.csv` based on the values observed. Below is the recommended data type mapping for each column in the dataset:

Invoice Number (e.g., 536378 , 536366): Numeric. Suggested data type: `INTEGER` .
Stock Code (e.g., 22888 , 22883 , 88123A): Alphanumeric that can include both numbers and characters. Suggested data type: `VARCHAR` or `TEXT` .
Description (e.g., JUMBO BAG PINK POLKA DOT , HAND WARMER UNION JACK): Textual information. Suggested data type: `TEXT` or `VARCHAR` with a sufficient length (e.g., `VARCHAR(255)`).
Quantity (e.g., 18 , 6 , 32): Numeric. Suggested data type: `INTEGER` .
Invoice Date (e.g., 2023-12-01 09:37 , 2023-12-01 08:28): Date-time information. Suggested data type: `DATETIME` or `TIMESTAMP` .
Price (e.g., 1.99 , 2.59 , 3.19): Real numbers with decimal places. Suggested data type: `DECIMAL` or `FLOAT` with precision (e.g., `DECIMAL(10, 2)`).
Customer ID (e.g., 14689 , 17889 , 13947): Numeric. Suggested data type: `INTEGER` .
Country (e.g., UNITED KINGDOM , FRANCE): Textual information. Suggested data type: `TEXT` or `VARCHAR` with a sufficient length (e.g., `VARCHAR(100)`).

Schema Using Recommended Data Types:

Ask a design question about your data schemas...